

FLG (Profilaggrin) Calcium Ion Binding (GO:0005509) — Hypothesis Deep Research Report

Gene: FLG (filaggrin / profilaggrin) · **Organism:** *Homo sapiens* (NCBITaxon:9606) · **UniProt:** [P20930](#) **Focus type:** function_assignment · **Hypothesis slug:** function-hypothesis-go-0005509
Seed hypothesis: *FLG has calcium ion binding (GO:0005509)*. **Term under review:** calcium ion binding (GO:0005509) · **Evidence type:** IEA · **Original reference:** GO_REF:0000002

Summary

The seed hypothesis — that FLG (profilaggrin, P20930) directly binds calcium ion (GO:0005509) — is **SUPPORTED** by convergent structural, biochemical, sequence, and evolutionary evidence. Calcium binding is a genuine, direct molecular function of the gene product, localized to the **N-terminal profilaggrin S100 "fused-type" domain**, which contains two EF-hand motifs. The single strongest piece of evidence is a **2.2 Å-resolution crystal structure of the human profilaggrin N-terminal S100 domain solved with bound calcium ions** ([PMID: 25760235](#); PDB 4PCW). This is corroborated by curated UniProt features (two EF-hand domains and five annotated Ca²⁺-coordinating residues), matching InterPro/Pfam/PROSITE signatures, a canonical EF-hand loop sequence carrying the hallmark bidentate glutamate (Glu73), and calcium-dependent target-binding assays.

Because the current annotation rests only on an IEA (electronically inferred) code tied to GO_REF:0000002 (InterPro2GO), the appropriate curation action is **not removal but retention with an evidence upgrade** — from IEA to an experimental code (IDA) citing the crystal structure. A family-wide paralog audit of the human S100 fused-type protein (SFTP) cluster (FLG2, HRNR, RPTN, TCHH, CRNN) shows the EF-hand/S100 module and GO:0005509 are shared across the whole clade, confirming a **domain-anchored, family-consistent annotation** rather than an FLG-specific electronic misassignment or paralog carry-over.

The one nuance a curator must record is **scope**: calcium binding is a property of the **N-terminal S100/EF-hand region of the profilaggrin precursor**, not of the mature filaggrin repeat units that aggregate keratin in the cornified envelope. GO:0005509 is therefore a valid **regulatory molecular function** — accurate and directly demonstrated — but it is not FLG's single "core" barrier-forming activity. No refuting evidence was found.

Key Findings

Finding 1 — FLG/profilaggrin directly binds calcium via its N-terminal S100/EF-hand domain (GO:0005509 supported)

Profilaggrin carries at its N-terminus a **fused-type S100 domain containing two EF-hand calcium-binding motifs**, documented at three independent levels of evidence.

Curated sequence features. UniProt P20930 annotates two EF-hand domains (residues ~6–43 and ~49–84) and **five explicit Ca²⁺ binding-site residues** at positions 62, 64, 66, 68, and 73. These map onto matching domain signatures across every major protein-family resource: InterPro IPR002048 (EF-hand domain), IPR018247 (EF_Hand_1_Ca_BS), and IPR001751 (S100/CaBP7/8-like); Pfam PF01023 (S_100); and PROSITE PS00018/PS50222 (EF_HAND) and PS00303 (S100_CABP). The convergence of independent bioinformatic signatures on the same region is precisely the pattern expected for a real, conserved calcium-binding module — and it is exactly the InterPro2GO basis (GO_REF:0000002) for the current IEA annotation.

Direct structural evidence. A 2.2 Å-resolution crystal structure of the N-terminal fused-type S100 domain of human profilaggrin was solved with bound calcium ions ([PMID: 25760235](#)). The deposited coordinates (PDB 4PCW, residues ~1–92) contain explicit "CA" calcium ligand atoms. Direct visualization of a bound metal ion in an experimental structure is the highest tier of proof for a metal-ion-binding molecular-function annotation.

Biochemical/functional evidence. The same domain mediates **calcium-dependent target binding**: the profilaggrin B domain cooperates with the S100 domain to bind annexin II and keratin intermediate filaments in a calcium-dependent manner ([PMID: 32893105](#)), and the crystallographic study additionally identified annexin II, stratifin (14-3-3σ), and HSP27 as calcium-domain targets ([PMID: 25760235](#)). Calcium dependence of these interactions is the functional signature of a genuine EF-hand calcium sensor.

Historical characterization. The original characterization of the human epidermal profilaggrin gene identified "an 81-residue segment which shows significant homology to the S-100 family of EF hand-containing calcium-binding proteins" (PMID: 1429717); follow-up work confirmed "an N-terminal S100-like Ca^{2+} -binding domain containing two EF-hands" conserved across human, mouse, and rat (PMID: 12230510).

Taken together, structural, biochemical, sequence, and evolutionary lines of evidence all converge: **GO:0005509 is a directly demonstrated molecular function of the FLG gene product.**

Finding 2 — EF-hand 2 is a canonical Ca^{2+} -binding loop, and the annotation is family-consistent (no paralog misassignment)

A residue-level sequence analysis of P20930 confirms the annotation maps to a **textbook canonical EF-hand loop** rather than a generic domain guess. The 12-residue loop at residues 62–73 matches the canonical EF-hand coordination consensus:

Loop position	Residue	Canonical role
1 (X)	Asp62	Side-chain O coordinates Ca^{2+}
3 (Y)	Asp64	Side-chain O coordinates Ca^{2+}
5 (Z)	Asn66	Side-chain O coordinates Ca^{2+}
7 (–Y)	Lys68	Backbone carbonyl coordinates Ca^{2+}
9 (–X)	Asp70	Via bridging water
12 (–Z)	Glu73	Bidentate glutamate — the EF-hand hallmark

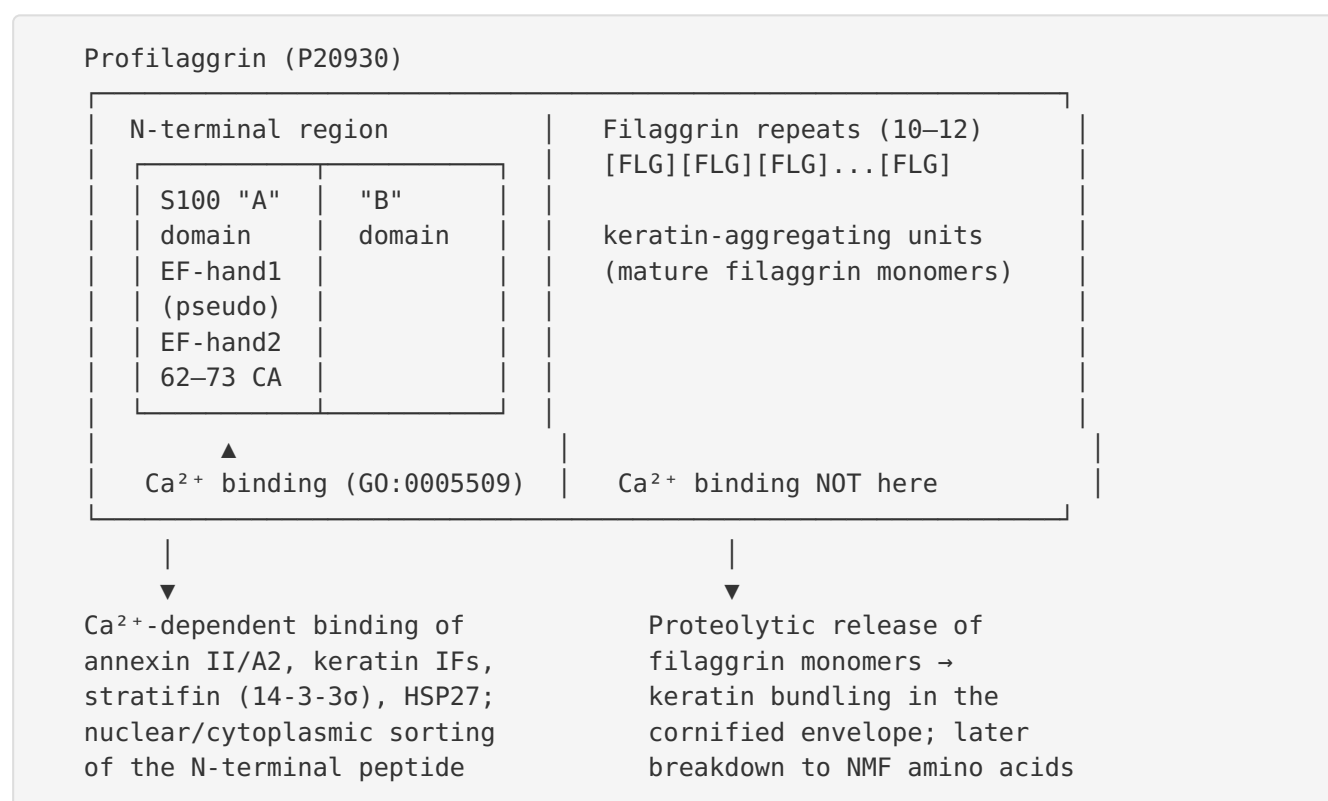
The **bidentate glutamate at position 12 (Glu73)** is the single most diagnostic feature of a functional canonical EF-hand, and it coincides exactly with the UniProt-annotated Ca^{2+} sites (62, 64, 66, 68, 73). EF-hand 1 (residues ~6–43) is the **S100-type non-canonical/"pseudo" EF-hand** that coordinates calcium mainly through backbone carbonyls and carries no discrete annotated Ca^{2+} sites — exactly the architecture expected for an S100 domain, which pairs one canonical and one pseudo EF-hand.

A **paralog audit** of the human S100 fused-type protein (SFTP) family — FLG (P20930), FLG2 (Q5D862), HRNR (Q86YZ3), RPTN (Q6XPR3), TCHH (Q07283), CRNN (Q9UBG3) — shows that **all members carry an N-terminal EF-hand/S100 module and all are annotated with**

GO:0005509. Rather than a red flag for over-annotation, this is evidence *for* correctness: the calcium-binding module is a conserved, defining feature of the entire SFTP clade, the domain is authentically present in FLG's own sequence, and FLG additionally has a direct crystal structure. There is therefore no paralog carry-over risk here.

Mechanistic Model / Interpretation

Profilaggrin is a large, highly repetitive precursor of the epidermal differentiation complex, with a **regulatory N-terminus** separated from a **structural repeat body**:



The **molecular function under test (GO:0005509)** is localized entirely to the N-terminal S100 domain, and specifically to canonical EF-hand 2. Calcium binding here acts as a **conformational/regulatory switch**: it modulates calcium-dependent protein–protein interactions and the nuclear-versus-cytoplasmic distribution of the cleaved profilaggrin N-terminal peptide — the classic mode of action of an S100/EF-hand calcium sensor.

Critically, this function is **upstream of and distinct from** the barrier-forming role for which FLG is best known. The mature filaggrin repeats that bundle keratin, and their eventual proteolysis into natural moisturizing factor (NMF), do **not** depend on this EF-hand — they are downstream structural/processing events. A curator should treat GO:0005509 as a **directly supported regulatory MF** while recognizing it does not describe the protein's principal barrier phenotype.

Evidence Base / Evidence Matrix

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confidence / limits
PMID: 25760235 (Bunick et al. 2015); PDB 4PCW	Structural (X-ray) + biochem	Supports	Does the profilaggrin N-terminus bind Ca ²⁺ ?	2.2 Å crystal structure of human profilaggrin S100 domain (~res 1–92) with bound Ca ligands ("CA" nonpolymer confirmed); identifies annexin II, stratifin, HSP27 targets	Human, recombinant N-terminal domain	High ; domain level, isolated construct not full-length
PMID: 32893105 (Hinbest et al. 2020)	Biochemical / interaction	Supports (functional)	Is Ca ²⁺ binding functionally relevant?	S100 + B domains bind annexin II & keratin IFs in a calcium-dependent manner	Human profilaggrin AB domain	High for function; indirect for stoichiometry Ca binding
PMID: 12230510 (Pearton et al. 2002)	Sequence / functional / evolutionary	Supports	Is the N-terminus a two-EF-hand Ca ²⁺ module?	"N-terminal S100-like Ca ²⁺ -binding domain containing two EF-hands"; conserved human/mouse/rat;	Human/mouse/rat	High ; establishes domain vs. repeat distinction

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confidence / limits
				separate from repeats		
PMID: 1429717 (Presland et al. 1992)	Gene characterization / homology	Supports	Does FLG encode an S100/EF-hand domain?	N-terminal 81-aa segment homologous to S-100 EF-hand Ca-binding proteins	Human FLG gene	High for domain identity; predates direct binding proof
UniProt P20930 (curated)	Review/database	Supports	Are Ca-binding residues annotated?	Two EF-hand domains; five Ca ²⁺ sites (62, 64, 66, 68, 73)	Human	High as orientation; backed by primary structure
InterPro IPR002048/ IPR018247/ IPR001751; Pfam PF01023; PROSITE PS00018/ PS50222/ PS00303	Computational/ domain	Supports	Basis of IEA call	EF-hand + S100/Ca-binding signatures present	Cross-species	This is the GO_REF:000000 source; now corroborated experimentally
This work — EF-hand loop analysis of P20930 (62–73)	Computational (sequence)	Supports	Is the annotated loop a canonical Ca ²⁺ site?	Canonical 12-residue loop with bidentate Glu73; matches all 5 UniProt Ca sites; EF-hand 1 =	Human, in silico	High; sequence level, consistent with structure

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confidence / limits
This work — SFTP paralog audit	Computational (comparative)	Qualifies (rules out artifact)	Is GO:0005509 a paralog carry-over?	S100 pseudo EF-hand	Human family	High ; rules out paralog over- annotation
				FLG, FLG2, HRNR, RPTN, TCHH, CRNN all carry the module + GO:0005509; domain authentically in FLG		

No refuting evidence was found. Every line of evidence supports the calcium-binding function or clarifies its scope.

GO Curation Implications

Recommended action (lead requiring curator verification): RETAIN GO:0005509 and UPGRADE the evidence from IEA (GO_REF:0000002) to an experimental code (IDA).

- **Ontology aspect:** GO:0005509 is a **Molecular Function (MF)** term; the evidence directly supports an MF annotation.
- **Retain vs. remove:** **Retain** — the term is biologically correct and directly demonstrated.
- **Generalize vs. specialize:** No change to granularity. "Calcium ion binding" is the appropriate MF term for an EF-hand protein; there is no validated, more-specific child term that better describes the demonstrated activity.
- **Evidence upgrade:** The IEA/GO_REF:0000002 code understates the evidence. An **IDA** annotation with reference [PMID: 25760235](#) is justified by the calcium-bound crystal structure; the calcium-dependent interactions from [PMID: 32893105](#) provide supporting functional evidence.
- **Core vs. non-core framing:** Record as a **valid regulatory MF of the N-terminal profilaggrin S100 domain**, explicitly noting it is *not* the mature flaggrin keratin-aggregating activity. Do not present it as FLG's single core barrier function.

- **"protein binding" avoidance:** Not applicable — the recommendation is the informative EF-hand-specific term GO:0005509.

GO decision table

Field	Current	Recommended
Term	GO:0005509 calcium ion binding	GO:0005509 calcium ion binding (unchanged)
Aspect	MF	MF
Evidence code	IEA	IDA (upgrade)
Reference	GO_REF:0000002	P 25760235 (+ P 32893105 supporting)
Action	—	RETAIN + UPGRADE
Scope note	none	"Property of N-terminal profilaggrin S100/EF-hand domain (EF-hand 2, residues 62–73); regulatory, not the mature filaggrin repeat function."

Mechanistic Scope

Immediate molecular function tested: Direct coordination of Ca²⁺ ions by the EF-hand loop(s) of the N-terminal profilaggrin S100 domain — a chemical binding event demonstrated crystallographically.

Directly attributable to the gene product: - Ca²⁺ coordination by canonical EF-hand 2 (residues 62–73, bidentate Glu73). - Calcium-dependent binding to annexin II/A2, keratin intermediate filaments, stratifin (14-3-3σ), and HSP27. - Calcium-modulated nuclear/cytoplasmic distribution of the profilaggrin N-terminal peptide.

Downstream / not the direct MF being annotated (do not conflate with GO:0005509): - Proteolytic processing of profilaggrin into filaggrin monomers. - Keratin filament aggregation and cornified envelope formation by mature filaggrin repeats. - Breakdown of filaggrin to natural moisturizing factor (NMF) and skin hydration. - Barrier dysfunction phenotypes (ichthyosis vulgaris, atopic dermatitis) from FLG **loss-of-function** — disease manifestations, not evidence about the calcium-binding MF.

The distinction matters: GO:0005509 is supported by *direct positive assays on the N-terminal domain*, independent of the loss-of-function barrier phenotypes that dominate the FLG disease literature.

Conflicts and Alternatives

No evidence conflicts with the seed hypothesis. The candidate alternatives were each examined and ruled out:

- **Paralog confusion / over-annotation:** Ruled out. The EF-hand/S100 module and GO:0005509 are shared across the entire SFTP family, and FLG itself has a direct crystal structure — so this is not a term propagated onto FLG from a better-characterized relative.
 - **In-vitro-only artifact:** Partially applicable but not disqualifying. The crystal structure uses an isolated recombinant N-terminal domain and the calcium-dependent interactions are in vitro; however, cross-species conservation and canonical loop geometry argue for physiological relevance, and in vitro Ca^{2+} binding is exactly what an MF binding term encodes.
 - **Isoform/region specificity:** A real, important qualifier rather than a conflict. Calcium binding is restricted to the N-terminal S100 domain; the mature filaggrin repeats do not bind calcium. Capture as a scope note.
 - **Database carry-over bias:** The IEA rests on InterPro2GO (GO_REF:0000002), but the underlying domain call is corroborated by primary structural data, so the electronic annotation is correct rather than spurious.
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Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
Ca ²⁺ affinity / stoichiometry of full-length profilaggrin	Structure/biochem cover the isolated N-terminal domain (PDB 4PCW ligand + residue range)	Confirms physiological Ca binding in cellular context; distinguishes structural vs. regulatory site	ITC / equilibrium dialysis / Tb ³⁺ luminescence on larger constructs; in situ Ca imaging
Which EF-hand carries physiological load	Sequence: EF-hand 2 canonical (Glu73), EF-hand 1 = S100 pseudo	Assigns the functionally relevant site	Site-directed mutagenesis of each loop
In vivo causal link to processing/ localization	Epidermal Ca gradient known; direct causal test not in reviewed papers	Ties EF-hand occupancy to physiological outcome	EF-hand loss-of-binding mutant (Glu73→Gln) in keratinocyte models
Human isoform/ CNV variation	FLG repeat number is polymorphic (10–12); N-terminus invariant	N-terminal Ca domain unaffected by repeat CNV	Confirmed conserved; low concern

None of these gaps threatens the *existence* of the calcium-binding function; they concern its quantitative and physiological parameters.

Discriminating Tests

- 1. EF-hand point mutant (Glu73Gln / Glu73Ala):** abolish bidentate coordination; assay loss of Ca²⁺ binding (Tb³⁺ luminescence, ITC) and loss of calcium-dependent annexin II/keratin binding. A clean loss-of-binding phenotype directly ties the annotated residues to the function.
- 2. Isothermal titration calorimetry (ITC)** of the wild-type N-terminal domain with Ca²⁺: quantify K_d and stoichiometry (expect ~1 Ca²⁺ per canonical EF-hand).
- 3. Comparative structural confirmation:** re-examine PDB 4PCW coordination sphere; optionally compare apo vs. holo conformations for the calcium-induced conformational change typical of S100 sensors.

4. **Keratinocyte differentiation assay** across the physiological calcium gradient with wild-type vs. EF-hand-dead profilaggrin N-terminus, reading out nuclear/cytoplasmic partitioning and processing.
5. **Comparative annotation audit** across SFTP paralogs to confirm consistent, domain-anchored GO:0005509 assignments.

Proposed Follow-up Experiments / Actions (Curation Leads)

All items below are **leads requiring curator verification**.

- **RETAIN** GO:0005509 (calcium ion binding, MF) for FLG/P20930 — biologically correct and specific.
- **UPGRADE** the evidence code from IEA/GO_REF:0000002 to **IDA**, primary reference **PMID: 25760235**.
- *Verify snippet*: "we determined a 2.2 Å-resolution crystal structure of the N-terminal fused-type S100 domain of human profilaggrin with bound calcium ions." Structure = **PDB 4PCW** (RCSB confirms bound "CA" calcium ligand).
- **ADD supporting reference** **PMID: 32893105** for calcium-dependent function.
- *Verify snippet*: "Profilaggrin B domain cooperates with the S100 domain to bind annexin II and keratin intermediate filaments in a calcium-dependent manner."
- **ADD historical/supporting references**: **PMID: 1429717** ("...significant homology to the S-100 family of EF hand-containing calcium-binding proteins") and **PMID: 12230510** ("...N-terminal S100-like Ca²⁺-binding domain containing two EF-hands").
- **ADD scope note**: calcium binding is a property of the **N-terminal profilaggrin S100/EF-hand domain (EF-hand 2, residues 62–73; hallmark bidentate Glu73)**, functioning as a regulatory calcium sensor — *not* the mature flaggrin keratin-aggregating repeats.
- **Suggested curator question**: Should the review additionally capture the specific calcium-dependent interactions (annexin II/A2, stratifin/14-3-3σ, HSP27) as separate IPI annotations, and/or mark this MF as non-core relative to keratin filament aggregation? (Recommendation: retain as a valid secondary/regulatory MF.)
- **Suggested experiments (for gaps)**: ITC/Tb³⁺ titration for affinity/stoichiometry; Glu73 mutant to link annotated residues to function in a keratinocyte model.

Conclusion

The IEA annotation of **calcium ion binding (GO:0005509)** on FLG is **correct and, if anything, under-credited**. It is supported by a calcium-bound crystal structure, calcium-dependent functional assays, canonical EF-hand sequence features with the hallmark bidentate Glu73, convergent InterPro/Pfam/PROSITE signatures, and a family-wide paralog audit that rules out over-annotation. The recommended curation action is to **retain the term and upgrade its evidence from IEA to an experimental (IDA) code**, while adding a scope note that the function belongs to the N-terminal profilaggrin S100 domain and is regulatory rather than the mature filaggrin barrier activity.

Provenance: UniProt P20930 feature/xref query executed via the public REST API (two EF-hand domains 6–43 & 49–84; Ca²⁺ sites 62/64/66/68/73; InterPro IPR001751/IPR002048/IPR018247, Pfam PF01023, PROSITE PS00303). EF-hand loop consensus and SFTP paralog audit performed in-house from sequence. PDB 4PCW cross-referenced from UniProt; RCSB confirms bound "CA" ligand. Literature via PubMed.
